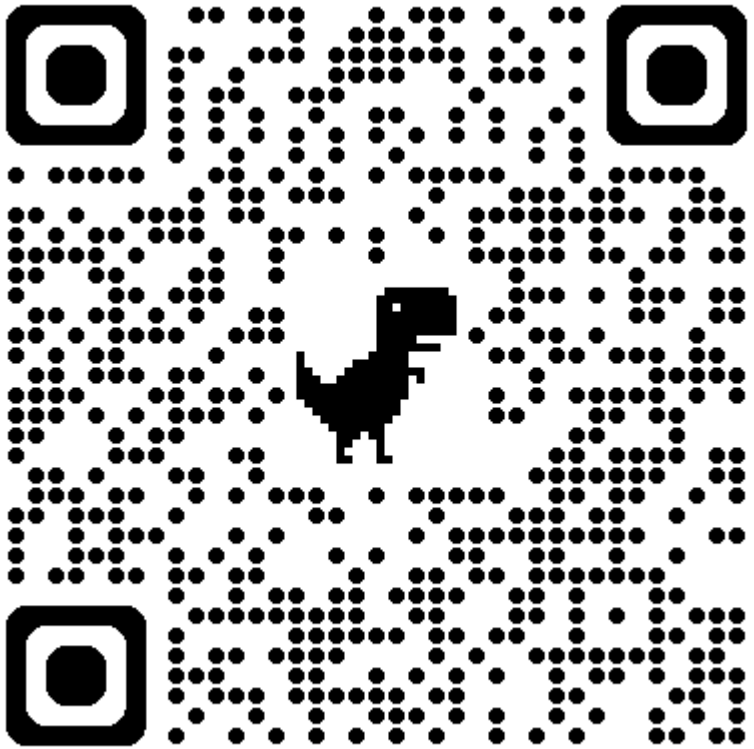
Introduction to Scientific Computing

ENGR1050 – Lecture 20

Fall 2026

Nat Trask

University of Pennsylvania



Introduction to AI-assisted programming

Penn
University of Pennsylvania

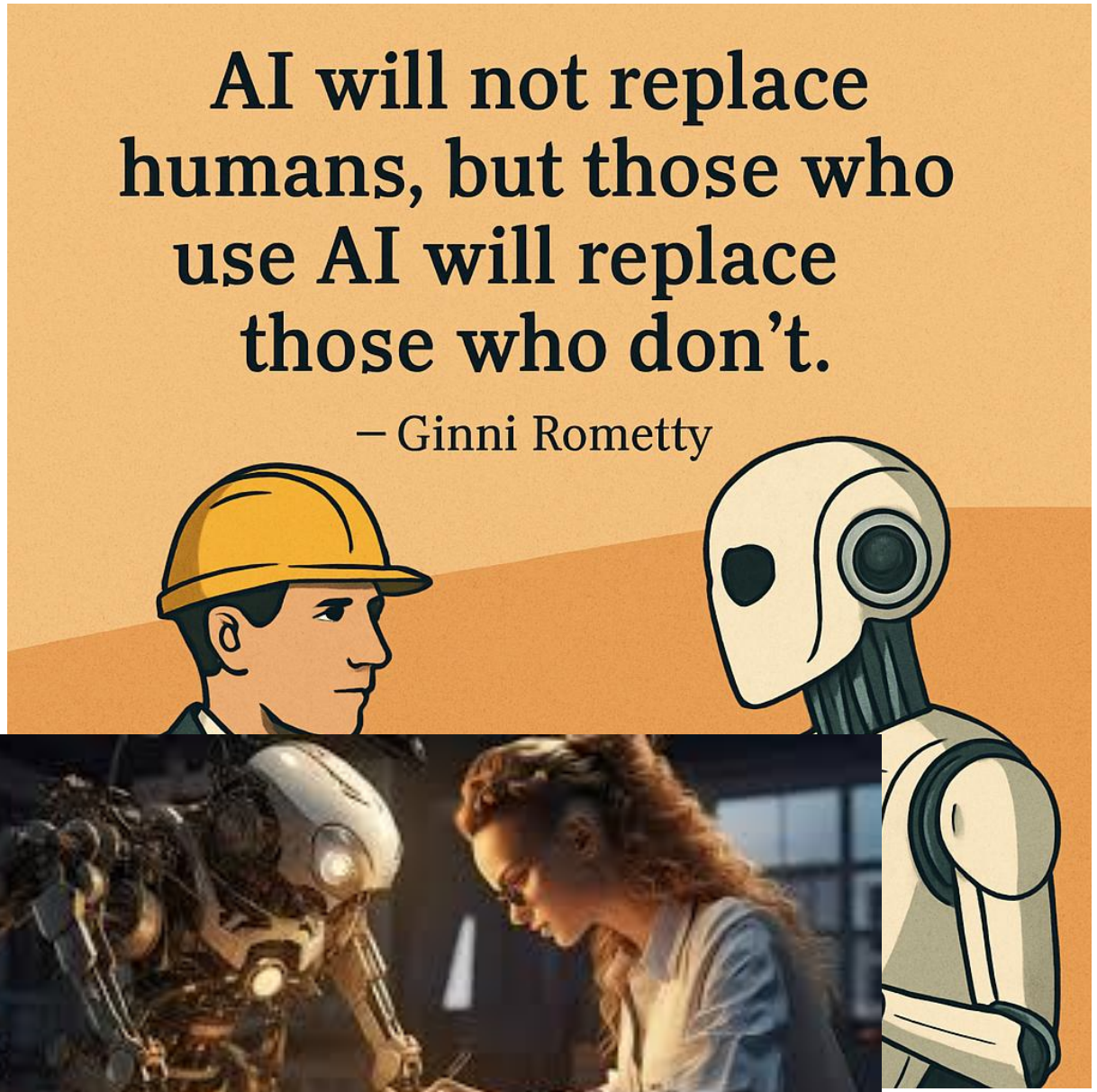
Intro to AI for programming unit

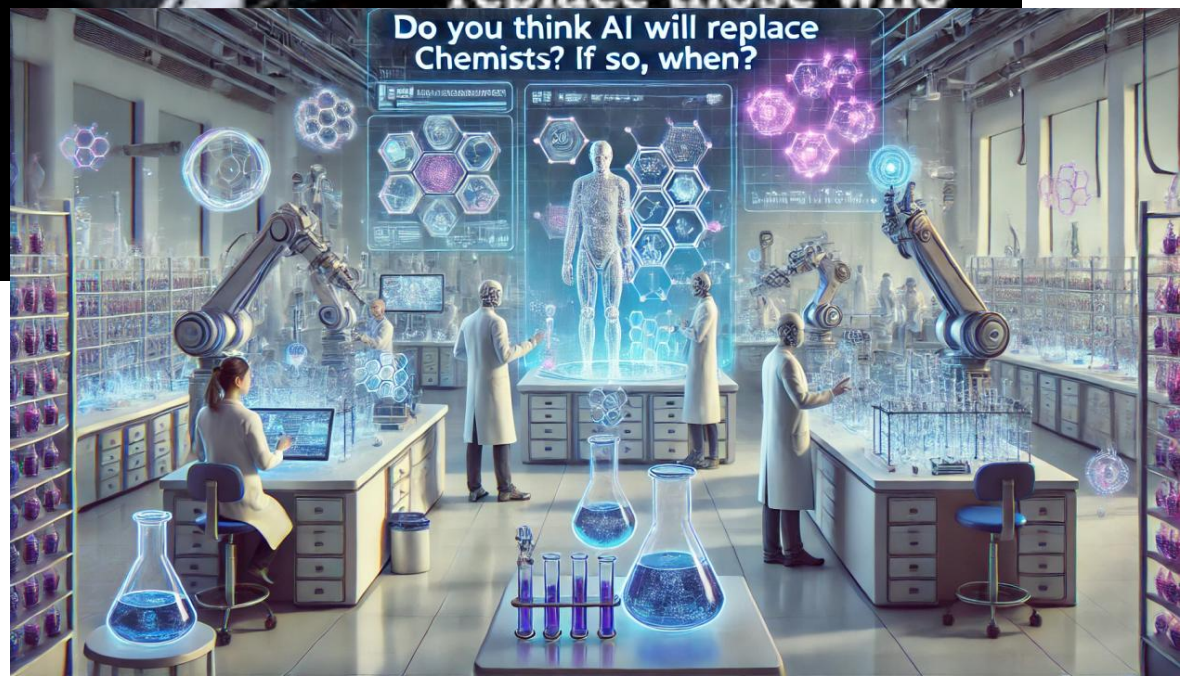
Writing code with AI

Using AI to learn new Python libraries

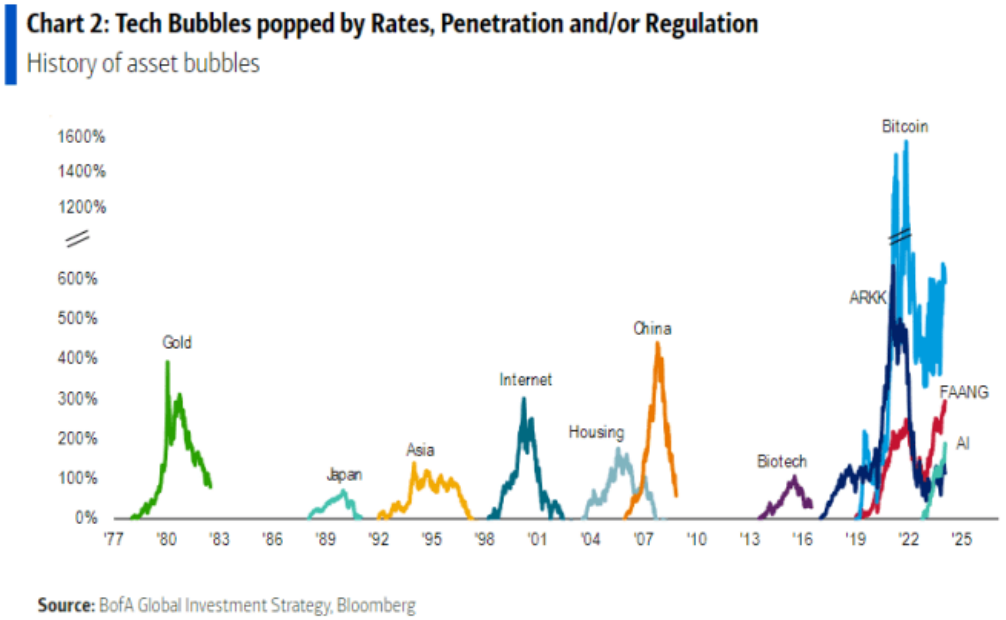
Running basic machine learning in Pytorch

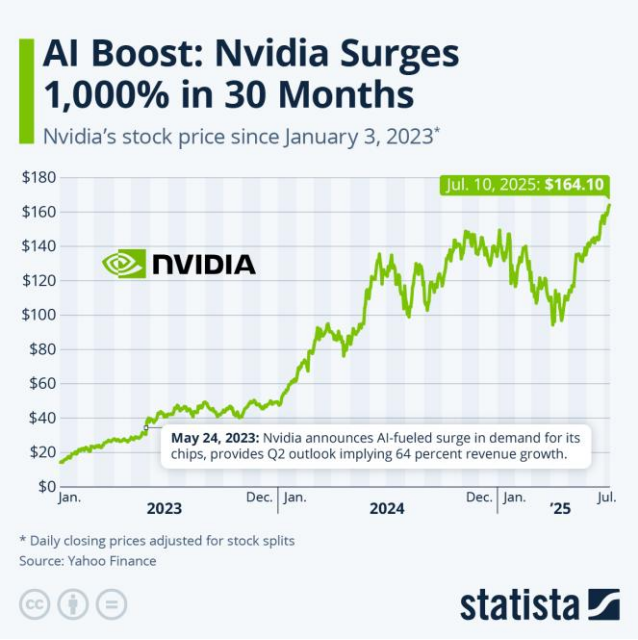

AI Learning Lesson
“AI won't replace humans — but humans who know how to use AI will replace those who


AI will not replace humans, but those who use AI will replace those who don’t.
– Ginni Rometty


Do you think AI will replace Chemists? If so, when?


Why AI Won't Replace Humans:
A SCIENTIFIC PERSPECTIVE
NADER AL MALLAH





There is massive uncertainty around AI

- **Tools are here that weren't here 2 years ago**
- **Tools here today might not be as good 2 years in the future**

What can we do to build strong fundamentals that future proof our skills as engineers?

Learning Goals

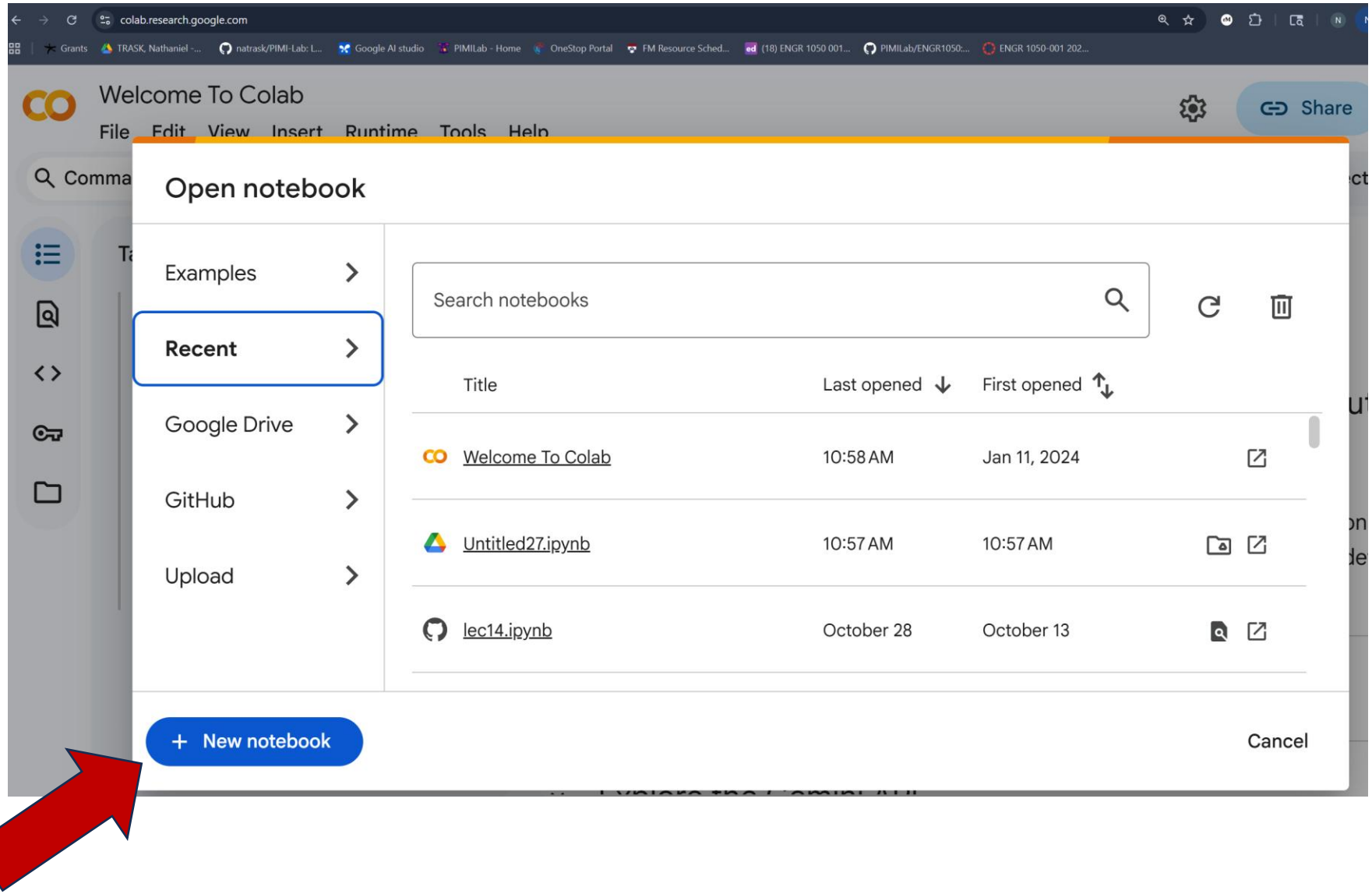
Literacy

Ethical Use

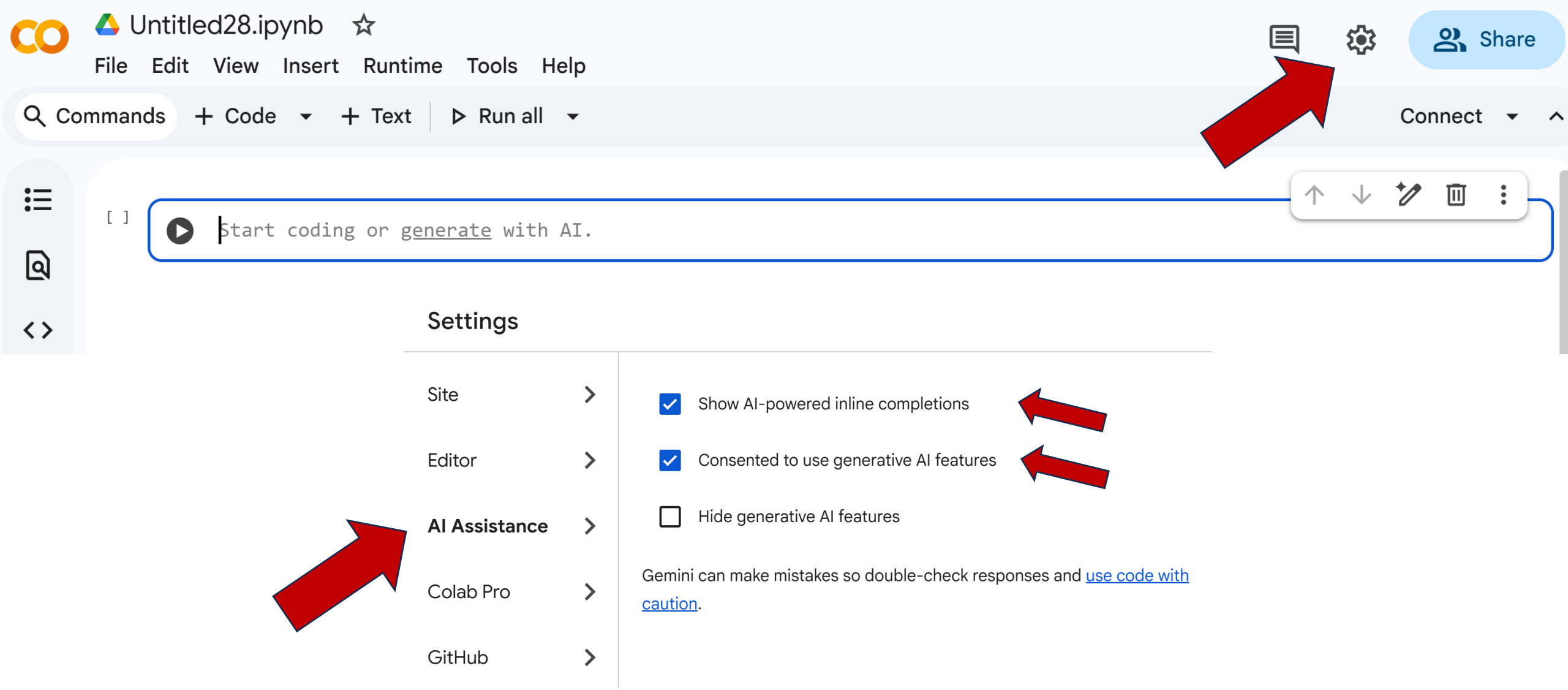
Technical Competency

Getting started with Gemini in Colab

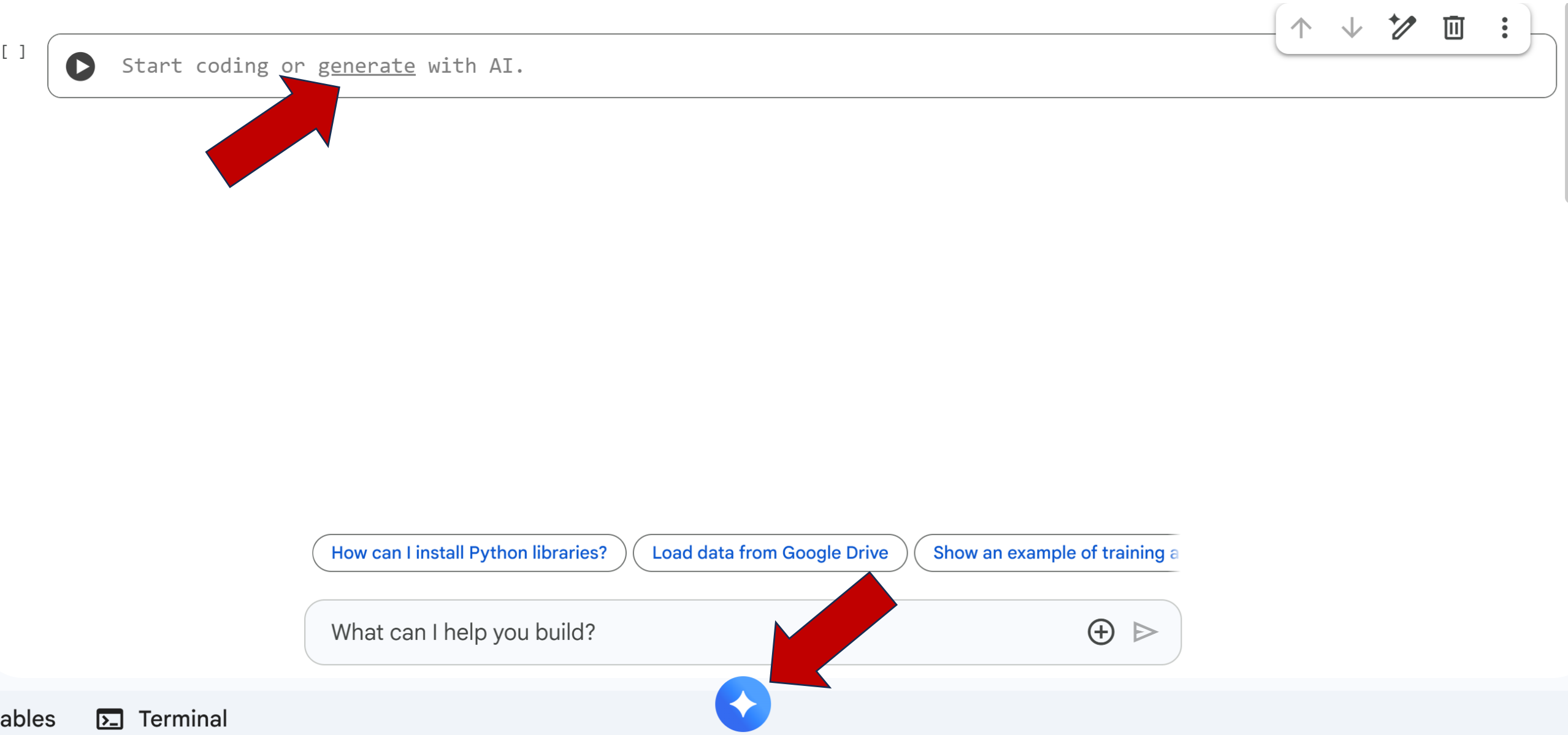
Open Colab > Click New notebook



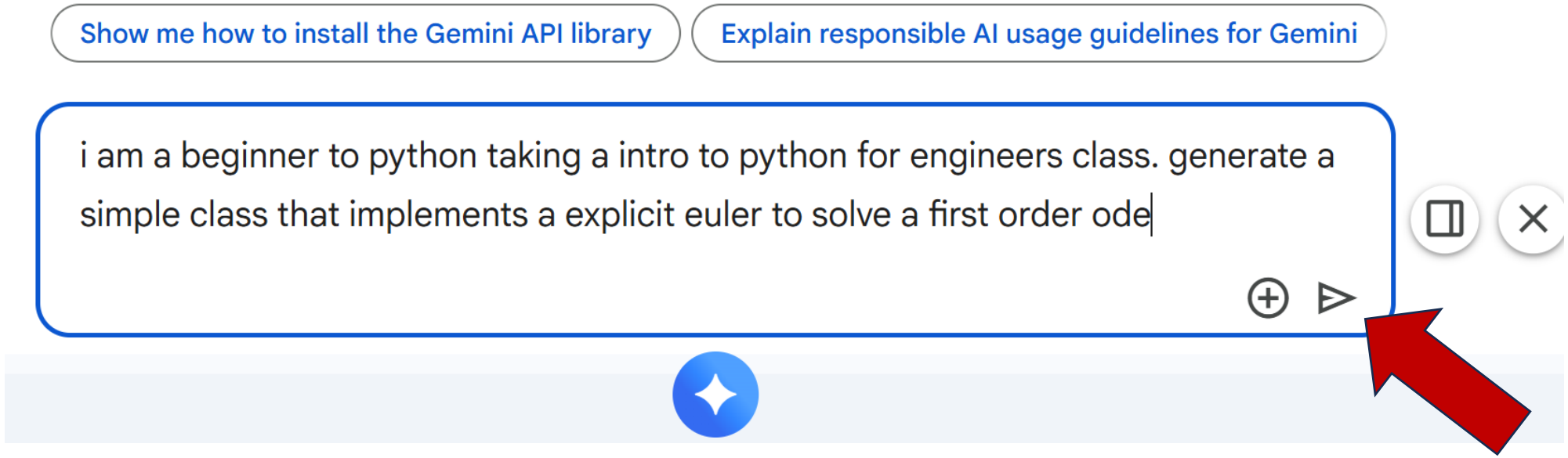
Click Settings icon > AI Assistance > Check inline completion + generative AI features



Two options: click generate with AI, or turn on Gemini chat dialog



A dialog will pop up that will write markdown+code for you



Explicit Euler Method

The explicit Euler method is one of the simplest numerical methods for approximating the solution of ordinary differential equations (ODEs) with a given initial value. It's a first-order method, meaning that the local error is proportional to the square of the step size, and the global error is proportional to the step size.

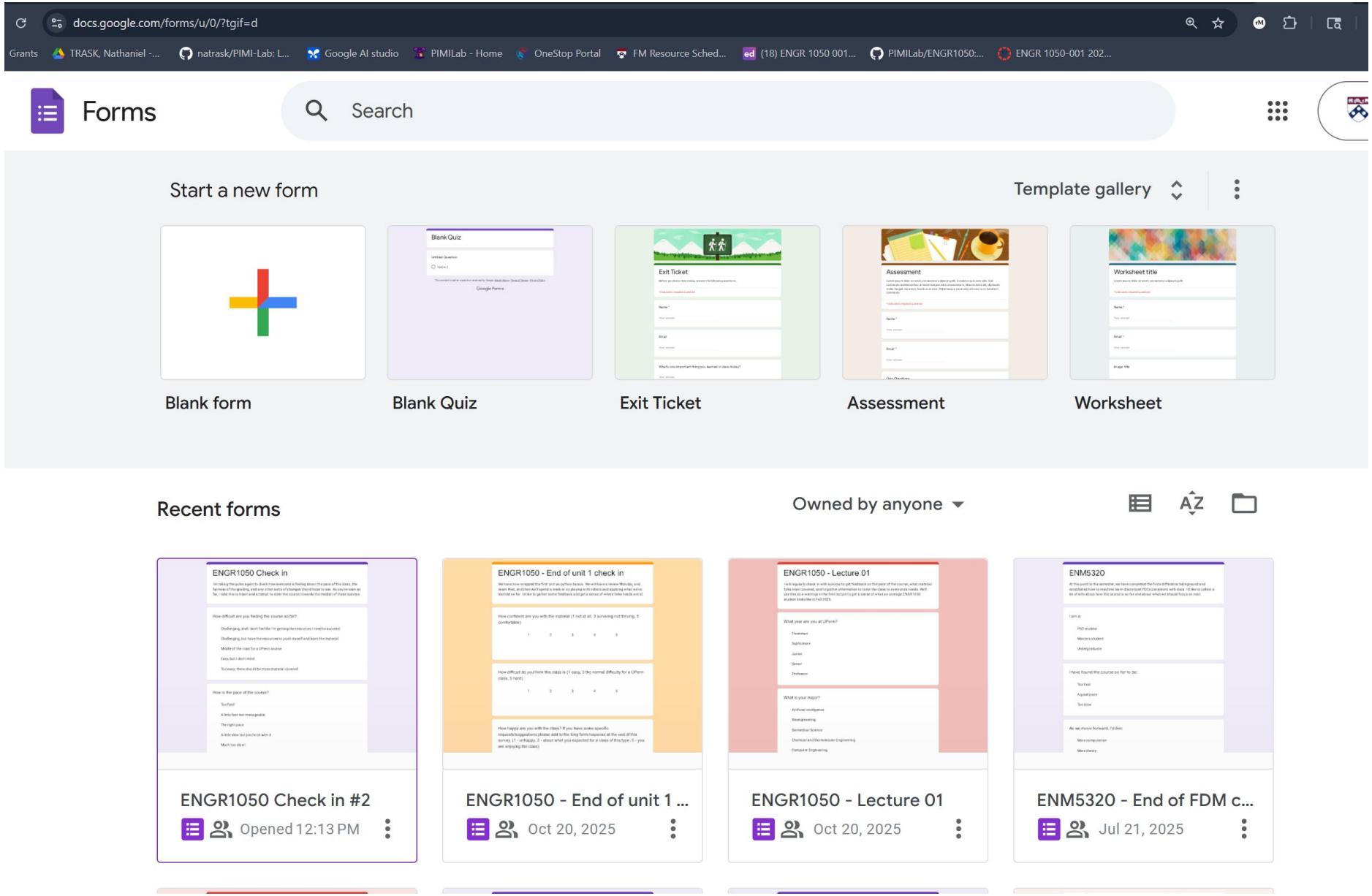
Given a first-order ODE of the form: $dy/dt = f(t, y)$ with an initial condition $y(t_0) = y_0$,

The explicit Euler method approximates the solution $y(t)$ at future time steps using the formula: $y_{i+1} = y_i + h * f(t_i, y_i)$

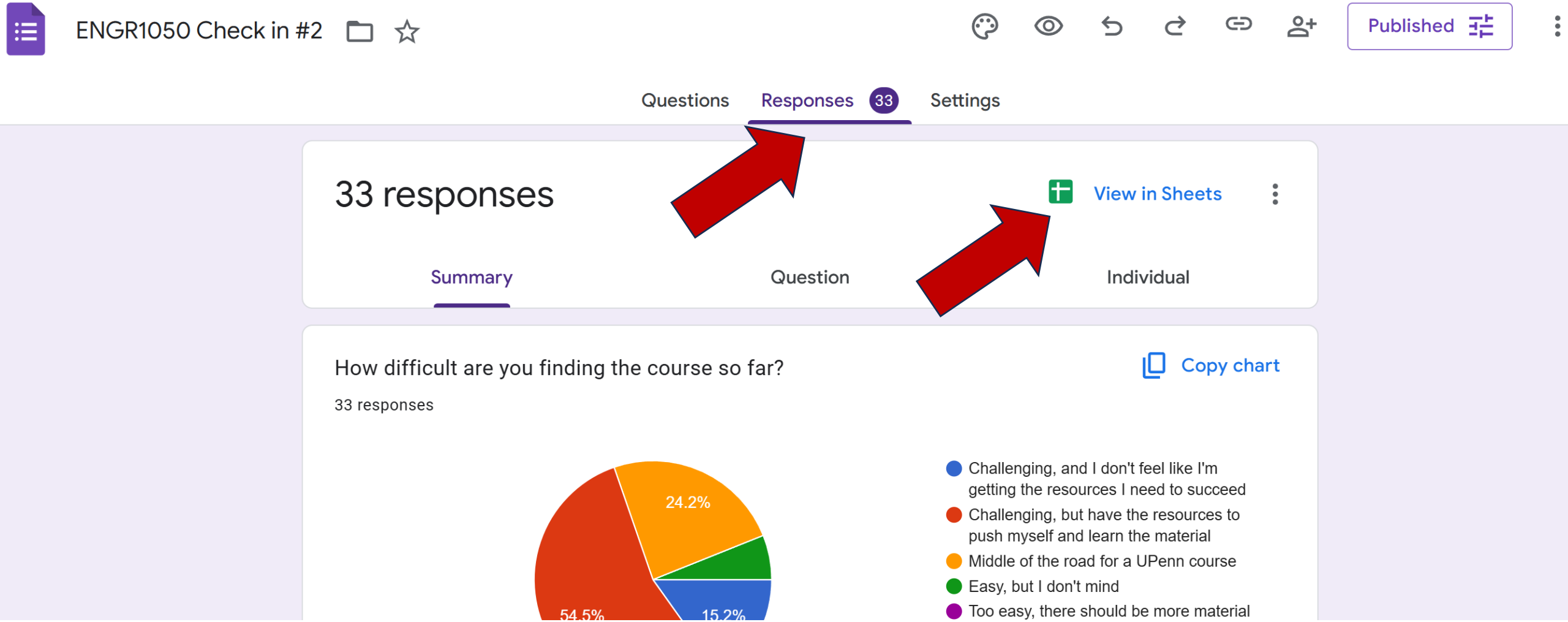
Where:

- y_i is the approximate value of the solution at time t_i.
- y_{i+1} is the approximate value of the solution at the next time step t_{i+1}.
- h (often denoted as dt) is the step size ($t_{i+1} - t_i$).
- $f(t_i, y_i)$ is the value of the derivative at t_i and y_i.

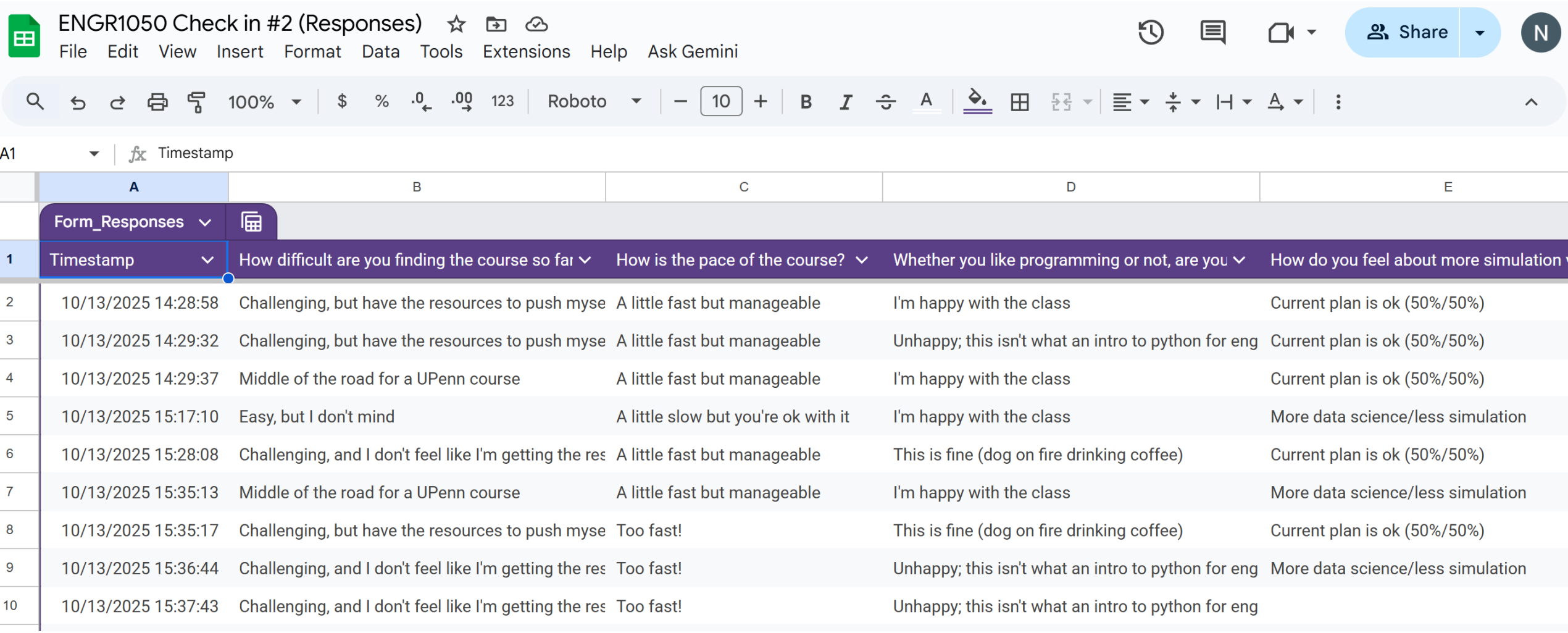
Todays task: Vibe-coding data collection



Responses > View in Sheets



Extracting data from this spreadsheet



ENGR1050 Check in #2 (Responses)

File Edit View Insert Format Data Tools Extensions Help Ask Gemini

Share

A1 | Timestamp

Form_Responses

	A	B	C	D	E
1	Timestamp	How difficult are you finding the course so far	How is the pace of the course?	Whether you like programming or not, are you	How do you feel about more simulation
2	10/13/2025 14:28:58	Challenging, but have the resources to push myse	A little fast but manageable	I'm happy with the class	Current plan is ok (50%/50%)
3	10/13/2025 14:29:32	Challenging, but have the resources to push myse	A little fast but manageable	Unhappy; this isn't what an intro to python for eng	Current plan is ok (50%/50%)
4	10/13/2025 14:29:37	Middle of the road for a UPenn course	A little fast but manageable	I'm happy with the class	Current plan is ok (50%/50%)
5	10/13/2025 15:17:10	Easy, but I don't mind	A little slow but you're ok with it	I'm happy with the class	More data science/less simulation
6	10/13/2025 15:28:08	Challenging, and I don't feel like I'm getting the res	A little fast but manageable	This is fine (dog on fire drinking coffee)	Current plan is ok (50%/50%)
7	10/13/2025 15:35:13	Middle of the road for a UPenn course	A little fast but manageable	I'm happy with the class	More data science/less simulation
8	10/13/2025 15:35:17	Challenging, but have the resources to push myse	Too fast!	This is fine (dog on fire drinking coffee)	Current plan is ok (50%/50%)
9	10/13/2025 15:36:44	Challenging, and I don't feel like I'm getting the res	Too fast!	Unhappy; this isn't what an intro to python for eng	More data science/less simulation
10	10/13/2025 15:37:43	Challenging, and I don't feel like I'm getting the res	Too fast!	Unhappy; this isn't what an intro to python for eng	

Next homework

